

Feedback on coursework

Designing consent for sharing

“We designed our consent forms and participant information with data sharing in mind, and made it a central part of both our ethics application and the study design more broadly ”

-> A quick technical note: from a legal perspective researchers are allowed to share personal data (under restricted/controlled access) without consent (!!)

because the GDPR has different legal basis, and ‘public task’ is such a legal basis. As publicly funded researchers, sharing is legitimate. Still, at minima, informing people that sharing happens is a good (and ethical) thing to do.

Provenance is at the data extraction level

“The extraction code may have bugs and live on the local machine of the person doing the extracting.”

“Any changes to the procedure or exclusion rules should also be dated and versioned.”

-> Reproducibility starts at data source extraction—not with the analysis-ready CSV. Any changes on how rules change gets documented (using .md and git makes this trivial).

Version control and complete workflows

“Any part of the analysis that lives outside the version-controlled base, however minor, is vulnerable to being lost or becoming inconsistent with the rest of the project.”

“An Excel calculation was deleted and subsequently moved into the scripted R workflow. “

-> Version control can help track these changes that inadvertently will happen

Change is hard

“It can be difficult to reach agreements to implement potential improvements in data structure that may affect the whole project and research performed on different sites.

-> Reproducibility is partly organizational, not merely individual.

Your work matters

“This would also help me know what happens to the data when I finish my PhD. Right now I do not have a clear answer to that last question.”

-> DMP allows to think beyond your current work

“It could be a good idea to store a copy on some cloud.”

-> Distinguish a personal copy from a versioned repository with metadata, licence and persistent identifier. (backup vs archive)

Can it be shared?

“Most of my data are inherently personal and therefore cannot be shared in its acquired form.”

-> There exist repositories to share personal data. Think e.g. about Statistics Denmark and the EU health data platform. Think e.g. about when you pay with your bank card, or take a plane. Personal data are shared all the time.

Deviations from pre-registration

“Since registration, several changes have been made to the data analysis plan, and these have been listed in the manuscript draft with a brief justification for each change.”

-> Deviations are allowed; concealing them is the problem.

Can you pre-register analyses of existing data?

“Analysis plan fixed before results — Not applicable. Everything has been exploratory so far—perhaps less applicable since I have existing data?”

“I mostly use publicly available datasets, and often I’ve already used them in previous projects. Therefore, the preregistration approach seems not to be adapted.”

-> You can still pre-register analyses based on publicly available data, see e.g. here: <https://aspredicted.org/cyq6-wnm7.pdf>

Programming vs Coding

“The code itself may be reproducible without the analytical decision being completely reproducible.”

“Rather than only recording that ten principal components were used, I could document how I evaluated different numbers of components and why ten were selected.”

-> Documenting hypotheses and reasoning than merely code comments

Do it yourself does not preclude collaboration

“I’m therefore now a bit reluctant when it comes to collaborations for code within my projects and prefer do it all myself.”

-> This means there is a single point of failure. A possible solution is sharing repositories, review, contribution rules and ownership of upstream processing.

Do it yourself does not preclude consistency

“I don’t follow a specific coding standard, as I’m writing research-only code.”

-> That’s totally fine, but consistency helps future researchers and the future yourself. A lightweight formatter, linter, tests and naming conventions provide substantial value. These conventions can be yours.

Cleanup after submission

“Often near deadlines or after reviews things become messier and part of Jupyter notebooks to be more efficient.”

-> Is your submitted work really reproducible if messy change happens after?